

RULE-BASED EXPERT SYSTEM

1. Introduction

An Expert System is a branch of Artificial Intelligence that mimics the decision-making ability of a human expert. This project diagnoses diseases based on symptoms entered by the user.

2. Objective

To develop a rule-based expert system that uses forward chaining to infer possible diseases from user-provided symptoms.

3. Software Requirements

- Python 3.x
- VS Code

4. Methodology

The system accepts symptoms as input and stores them as facts. A set of predefined IF-THEN rules is applied. When the conditions of a rule are satisfied, the system infers a disease and displays the reasoning process.

Example Rule:

IF fever AND cough THEN Flu

5. Implementation

The project uses:

- Facts Base (Symptoms)
- Rule Base (IF-THEN Rules)
- Inference Engine (Forward Chaining)

6. Sample Input

fever,cough

7. Sample Output

Rule Applied:

IF fever AND cough THEN Flu

Possible Diagnosis:

Flu

8. Conclusion

The Rule-Based Expert System successfully demonstrates AI reasoning using forward chaining and rule-based inference. The project is simple, efficient, and easy to understand.

9. Future Enhancements

- GUI using Tkinter
- Larger medical knowledge base
- Multiple disease prediction
- Web-based interface